

Infosys Virtual Internship 6.0

Project Title: A Machine Learning Model to Predict Fire Weather Index

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•Introduction

Forest fires pose a serious threat to the environment, wildlife, and human life. Early prediction of fire-prone conditions can significantly reduce damage by enabling preventive measures. The Fire Weather Index (FWI) is an important numerical indicator used to assess fire danger based on meteorological conditions such as temperature, humidity, wind speed, and rainfall.

This project focuses on building a machine learning-based regression model to predict the Fire Weather Index using historical weather data. The project is implemented as part of the Infosys Virtual Internship 6.0, covering dataset analysis, preprocessing, model building using Ridge Regression, performance evaluation, and deployment using a Flask web application.

Section 1: Dataset

1.1 Dataset Features

The dataset used in this project consists of meteorological and fire-related features. Each feature plays a crucial role in determining the fire weather conditions.

Day: Day of the month when the observation was recorded.

Month: Month of the year representing seasonal variation.

Year: Year of observation.

Temperature (°C): Daily average temperature, which strongly affects fire ignition probability.

RH (Relative Humidity): Measures moisture content in the air; lower values increase fire risk.

Ws (Wind Speed): Higher wind speed can accelerate fire spread

Rain: Amount of rainfall; rainfall generally reduces fire risk.

FFMC (Fine Fuel Moisture Code): Indicates moisture content of surface litter.

DMC (Duff Moisture Code): Represents moisture content of decomposing organic matter.

DC (Drought Code): Measures long-term drying effects.

ISI (Initial Spread Index): Indicates potential fire spread rate.

BUI (Build Up Index): Represents fuel availability.

FWI (Fire Weather Index): Target variable representing fire intensity.

Classes: Indicates fire occurrence severity.

Region: Encodes geographical region information.

1.2 Data Preprocessing

Introduction to Data Preprocessing

Data preprocessing is an essential step in machine learning to improve data quality and model performance. Raw datasets may contain missing values, duplicates, and features with different scales, which can negatively affect predictions.

In the Tempest FWI Predictor project, preprocessing ensures that the Fire Weather Index (FWI) dataset is clean, consistent, and suitable for Ridge Regression, which is sensitive to feature scaling.

Standardization:

Standardization is a feature scaling technique that transforms data so that it has:

Mean = 0

Standard Deviation = 1

Standardization is necessary because Ridge Regression is sensitive to feature scale. It ensures that all features contribute equally to the model.

Normalization:

Normalization helps scale values into a common range, improving numerical stability and convergence during training.

Normalization helps in:

- Improving numerical stability
- Preventing large feature values from influencing gradient updates
- Enhancing performance when features have highly skewed distributions

Section 2: Dataset Statistics

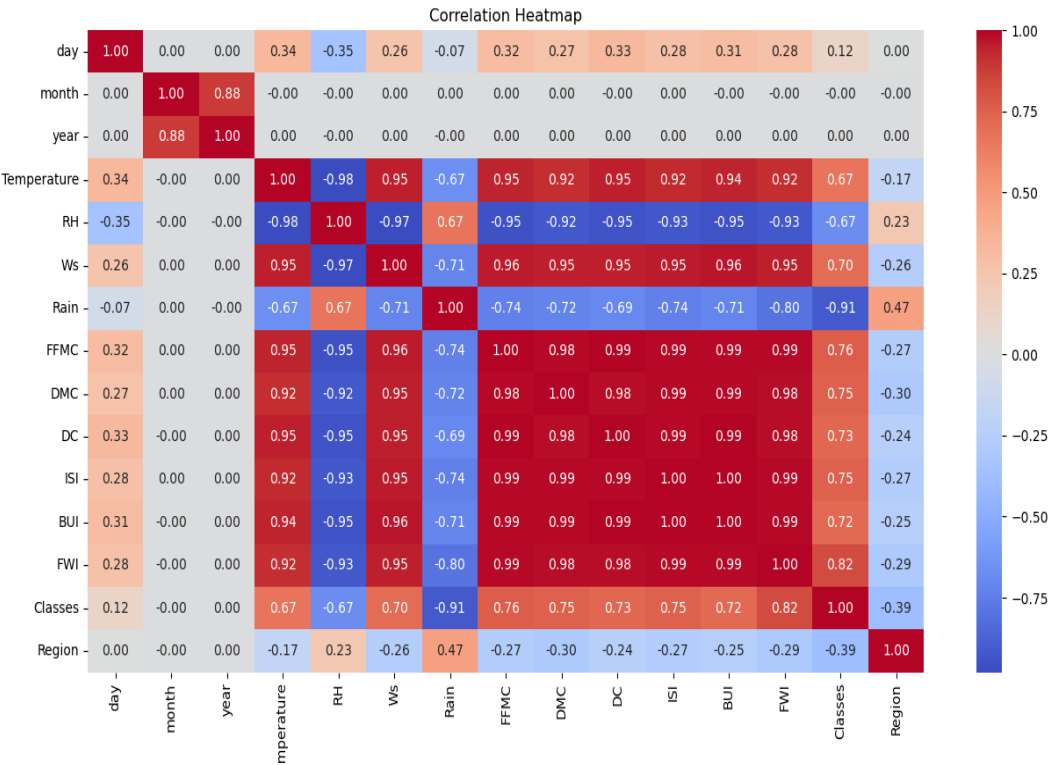
2.1 Correlation Analysis

Correlation analysis helps understand the relationship between different features and the target variable (FWI). A correlation heatmap is used to visualize these relationships.

Key observations from the correlation heatmap:

- FWI shows a strong positive correlation with FPMC, DMC, DC, ISI, and BUI.
- Temperature and wind speed are positively correlated with FWI.
- Rainfall has a negative correlation with FWI, indicating reduced fire risk during rainfall.

Some features exhibit multicollinearity, which justifies the use of Ridge Regression.



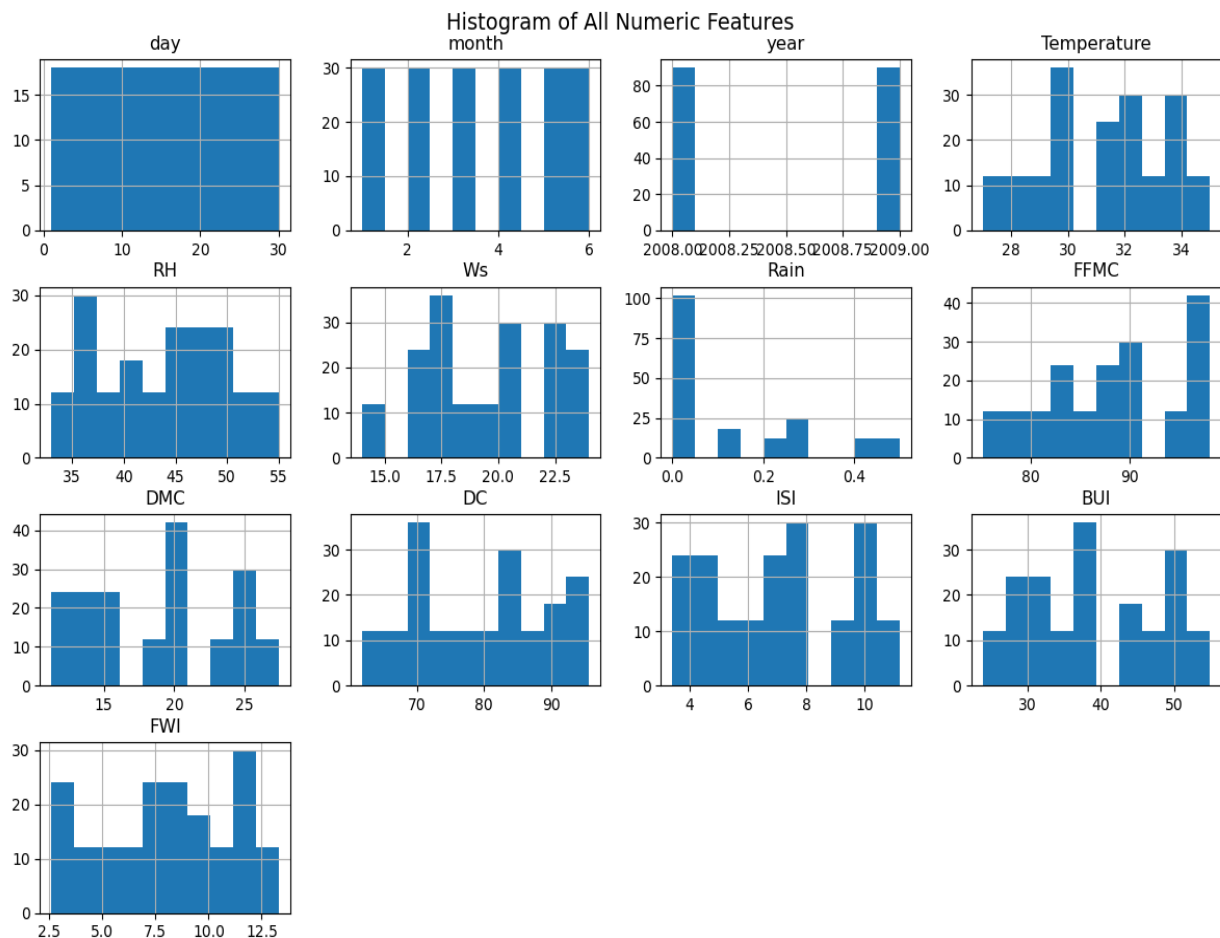
2.2 Histogram Analysis

Histograms are plotted for all numerical features to analyze their distributions.

Insights from histograms:

- Temperature values are concentrated within a moderate range, suitable for regression modeling.
- Rainfall shows a skewed distribution with many zero values.
- FWI values are spread across a reasonable range, indicating good variability for prediction.

Most features show non-uniform distributions, reinforcing the need for preprocessing.



Section 3: Ridge Regression

3.1 Ridge Regression and Alpha

Ridge Regression is a regularized version of linear regression that helps reduce overfitting by adding an L2 penalty term to the cost function.

The regularization strength is controlled by a hyperparameter called alpha (α):

- Small alpha $\rightarrow$ behaves like simple linear regression.
- Large alpha $\rightarrow$ stronger regularization, shrinking coefficients.

Proper alpha selection balances bias and variance.

3.2 Performance Analysis: MSE & MAE vs Alpha

To evaluate model performance, the following metrics are used:

- **MSE (Mean Squared Error):** Mean Squared Error measures the average of the squared differences between the actual and predicted values.

Interpretation:

- **Lower MSE** $\rightarrow$ better model performance
- **Higher MSE** $\rightarrow$ poor prediction accuracy

- **MAE (Mean Absolute Error):** Mean Absolute Error measures the average of the absolute differences between actual and predicted values

The model is trained with different alpha values, and error trends are observed.

Observations:

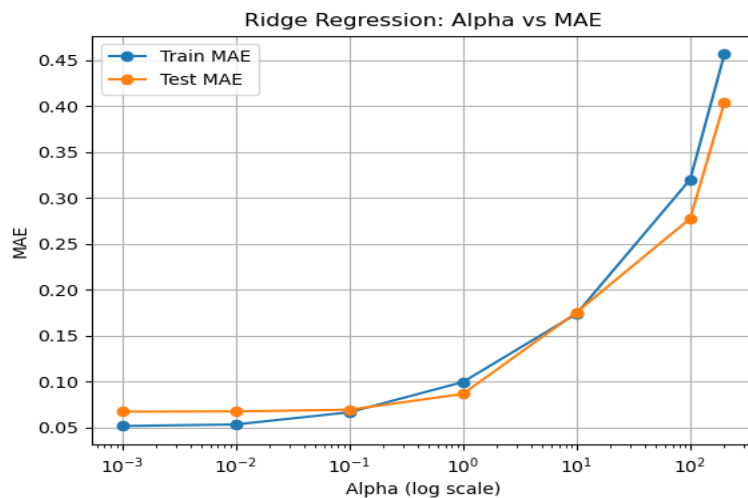
- For very small alpha values, training and testing errors are low.
- As alpha increases, both MSE and MAE increase due to excessive regularization.
- The optimal alpha lies in the lower range where test error is minimized.

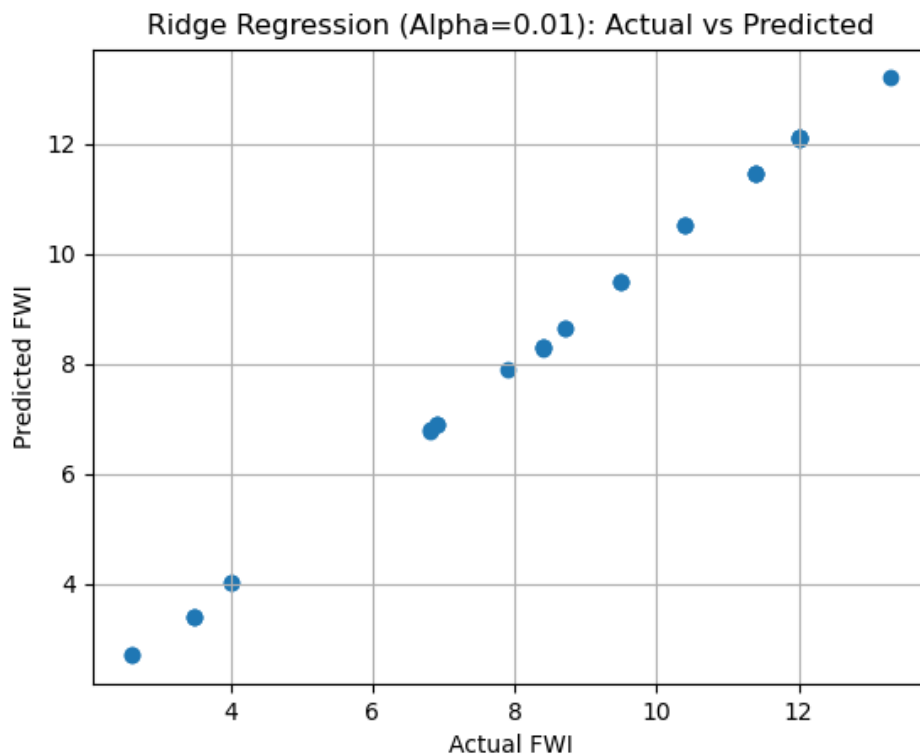
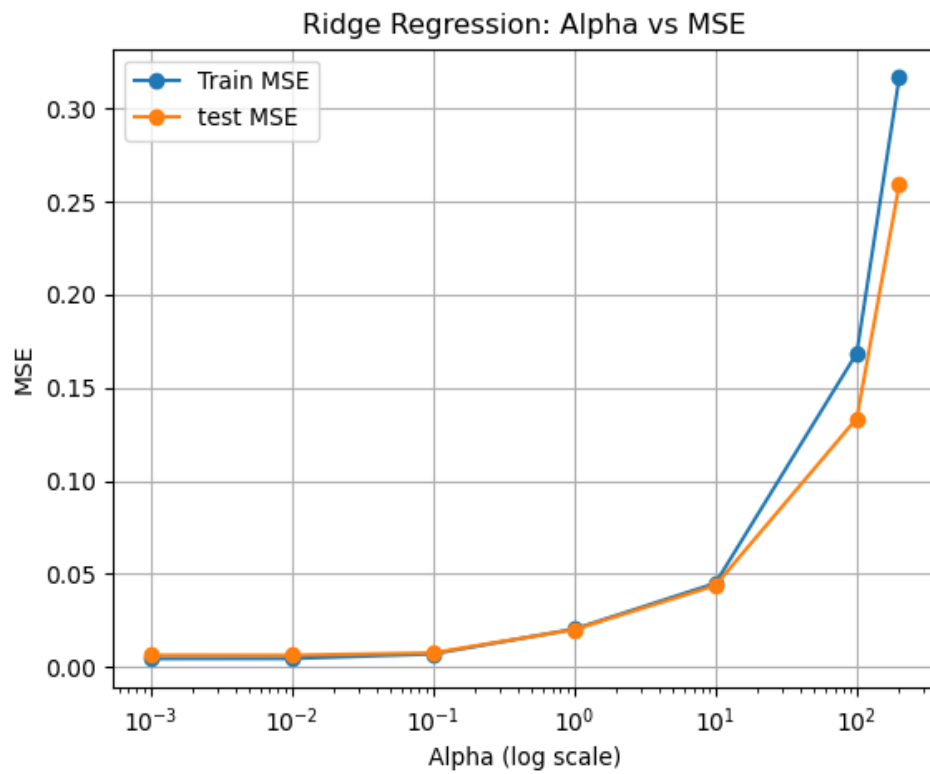
3) Hyperparameter Tuning – Alpha

Metrics tracked for multiple values of alpha:

Best alpha: 0.01 – chosen for the **lowest Test MSE** and balanced train-test performance.

Alpha	Train MSE	Test MSE	Train RMSE	Test RMSE	Train MAE	Test MAE	Train R ²	Test R ²
0.001	0.004444	0.006280	0.066667	0.079244	0.051540	0.067186	0.999555	0.999341
0.01	0.004506	0.006237	0.067128	0.078974	0.053221	0.067473	0.999549	0.999345
0.1	0.006976	0.007536	0.083524	0.086808	0.066502	0.069279	0.999302	0.999209
1.0	0.020310	0.020051	0.142513	0.141601	0.099442	0.086511	0.997969	0.997896
10.0	0.044952	0.043804	0.212020	0.209295	0.173814	0.175421	0.995504	0.995403
100.0	0.168569	0.133519	0.410571	0.365402	0.320241	0.277690	0.983141	0.985987
200.0	0.316688	0.259187	0.562750	0.509104	0.456688	0.404280	0.968326	0.972797





Section 4: Flask Application

4.1 Flask Backend Explanation

Flask is a lightweight Python web framework used to deploy the trained machine learning model.

Backend responsibilities:

- Load the trained Ridge Regression model.
- Accept user input through a web form.
- Apply preprocessing to input data.
- Generate FWI predictions.
- Display predicted results on the web interface.
- This enables real-time prediction of Fire Weather Index through a user-friendly interface.

4.2 Pseudo Code for Flask Application

START

Load trained Ridge Regression model

Initialize Flask application

Define route for home page

Display input form

Define route for prediction

Receive input values from user

Apply standardization/normalization

Pass input to trained model

Generate prediction

Return predicted FWI value

Run Flask server

END